Material-based Hertz two-sphere impact

LAMMPS dem02 hertz/material | Maximum compression, peak force and energy | 28 October 2026

PASS	4 of 4 runs pass the official 1% peak-energy check at 30.16 us.

Compression and force peaks

Run	Overlap (um)	Err. (%)	Force (N)	Err. (%)	Energy err. (%)	Official
CPU dt	274.112647	0.0000202	10696.885747	0.0000303	0.0000504	PASS
GPU dt	274.112647	0.0000202	10696.885747	0.0000303	0.0000504	PASS
CPU dt/2	274.112710	0.0000028	10696.889431	0.0000042	0.0000717	PASS
GPU dt/2	274.112710	0.0000028	10696.889431	0.0000042	0.0000717	PASS

Independent targets: overlap = 274.112702089 um; force = 10696.888983990 N. Peak values use every native step. Overlap/force 1% limits are additional diagnostics; the official checker asserts the sampled peak-energy relation only.

Direct material inputs and elastic mapping

Quantity	Official input / FunDEM configuration
Two spheres	Diameter 0.020 m; density 2800 kg/m^3; centers x = +/-0.0101 m; velocities vx = +/-10 m/s; zero spin
Normal law	hertz/material E 0.0 nu; E = 48 GPa; nu = 0.20. F = (4/3) E* sqrt(R*) delta^(3/2); R* = R/2.
FunDEM materials	Official E and nu are entered directly: E* = E / [2 (1 - nu^2)] = 25 GPa. Nonlinear stiffness; manual damping override off.
Friction / damping	Actual mu = 0.35 is retained. Zero slip/force/torque/spin makes tangential contact inactive. Restitution 1 gives zero normal damping; tangential damping = 0.
Time / full output	dt = 20 / 10 ns; 2508 / 5016 steps; same 50.16 us endpoint; native frames and CSV every step.
External forces	No wall or gravity; free boundaries. Both identical spheres remain inside the official box.

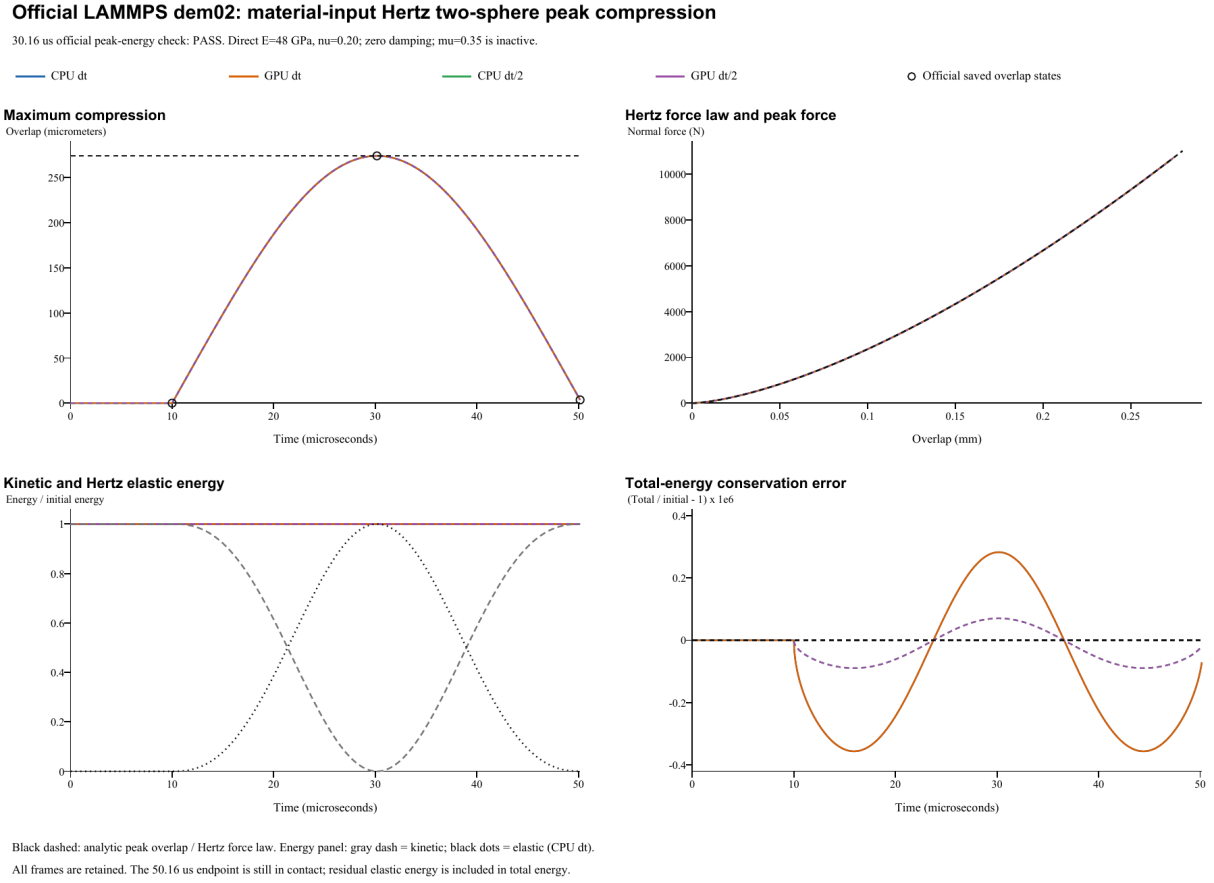
Official energy criterion at 30.16 us

E0 = (1/2) m* V^2 = 1.172861257 J; m* = m/2; V = 20 m/s.
U = (2/5) K delta^(5/2) = 0.4 F delta; K = (4/3) E* sqrt(R*).
The official checker compares E0 with 0.4 abs(F1) delta at step 1508, using 1% tolerance. It does not search for separate maxima or add residual kinetic energy at this near-peak instant.

Reference provenance

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob 7a5d55b4492162756a80aad7c965316a8068a0ec. LAMMPS was not rerun locally; YAML epsilon 1e-10 is not a universal cross-code tolerance.

Energy histories and retained evidence



Whole-event energy conservation

Run	Max. energy err. (%)	Final energy err. (%)	Rebound speed ratio
CPU dt	0.0000356	0.0000071	0.9999901564
GPU dt	0.0000356	0.0000071	0.9999901564
CPU dt/2	0.0000089	0.0000018	0.9999901824
GPU dt/2	0.0000089	0.0000018	0.9999901824

Whole-event maximum energy-error limit: a separately declared 1% diagnostic. E = kinetic energy of both spheres plus one pair elastic energy. Spin stays zero. The endpoint is still slightly compressed: its speed ratio is not an exact free-flight restitution test, and the residual elastic energy remains in E.

CPU/GPU velocities differ by at most 1.78e-15 m/s; maximum force-history difference is 0 N. Halving dt changes maximum whole-event energy error to 0.251065 of the base-step value. The fixed-time official PE error includes residual near-peak kinetic energy; it is not the global conservation error.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

2509 frames per base-step run; 5017 per half-step run. All 15,052 native frames, contacts, projects, every-step CSVs, audits and backend logs are retained.

Run manifest | Analysis JSON | Checkpoint CSV | Contact audit | Full native archive ZIP

Execution and scope

Real SimulationSession runs use linked FunDEM 0.4.3 CPU/CUDA native libraries, with no solver changes or parameter fitting. Byte identity with the installed GUI executable was not established.

This case checks direct E/nu mapping, elastic Hertz force/compression peaks, pair energy conservation and symmetric momentum. Nonzero mu is retained but inactive; dissipation, tangential response, spin and many-particle flows need other cases.